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**AgriPlus – IoT-Based Smart
Motor Monitoring & Control Platform****Software Requirements & Technical Architecture Document**

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1. PROJECT OVERVIEW**1.1 Introduction**

AgriPlus is an IoT-enabled Motor Automation System designed to provide realtime remote monitoring and control of agricultural water pumps.

The platform enables farmers and administrators to monitor electrical parameters, receive safety alerts, track energy usage, and manage devices securely through a mobile application.

1.2 Business Objectives

- Enable remote motor control from anywhere
- Prevent motor damage due to dry-run or overload
- Reduce travel time for farmers
- Provide real-time monitoring with minimal latency
- Deliver secure role-based system access
- Integrate e-commerce for agriculture accessories

1.3 Core Business Pillars (Specifications)

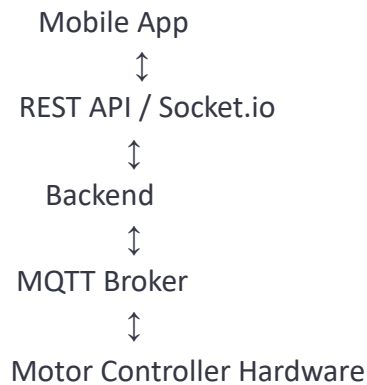
- **Real-Time Monitoring** – Live voltage, current, and power tracking
- **Remote Control** – Secure Start/Stop motor operations
- **Safety & Protection** – Dry-run and overload alerts
- **Energy Intelligence** – Consumption tracking and analytics
- **Buzzer indication** - Alerts for dry run / overload
- **Display the current / Voltage** – indicate the users 2 Phase / 3 Phase
- **Scheduler Start/ Stop** – Users can set the timing for Start / Stop
- **Motor Reverse Running** – If motor runs backward indicate to user.
- **OTA** – New firmware Upgradation
- **Read / Write Configuration** – Configuration details such as Motor device id, last updated firmware, Set Max/ Min Voltage and current and Power details.

2. SYSTEM ARCHITECTURE

2.1 Technology Stack

Layer	Technology
Mobile App	Flutter
Backend	Node.js, Express.js
Database	MongoDB (Mongoose ODM)
Real-Time Engine	Socket.io
IoT Protocol	MQTT
Caching	Redis
Authentication	JWT & Bcrypt
Payment Gateway	Razorpay

2.2 Conceptual Architecture Flow



2.3 Telemetry Data Flow

1. Hardware publishes data to MQTT topic agri/{serial_number}/telemetry
2. Backend validates:
 - ❖ Serial number
 - ❖ Frame structure
 - ❖ Device status
3. Processing Layer:
 - ❖ Safe JSON parsing
 - ❖ Voltage and current validation
 - ❖ Energy calculation
 - ❖ Fault detection
4. Distribution:

Socket.io → Mobile App Dashboard

FCM → Alert Notifications FCM → Mobile alerts

3. FUNCTIONAL REQUIREMENTS (SRS)

3.1 User Roles

Role	Description
Admin	System-level monitoring and management
User	Motor owner and operator

3.2 Core Functional Modules

3.2.1 Device Management

- Device registration
- Serial number mapping
- Device activation/deactivation
- Online/offline detection

3.2.2 Motor Control

- Remote Start motor
- Remote Stop motor
- Command acknowledgment tracking
- Real-time motor running status

3.2.3 Real-Time Tracking

- Live voltage display
- Current monitoring
- Power calculation
- RPM tracking
- Signal strength monitoring

3.2.4 Safety Features

- Dry run detection
- Overload alerts
- Low voltage alerts
- Push notification alerts

3.2.5 E-Commerce Module

- Product listing
- Cart management
- Order placement
- Razorpay payment integration
- Order lifecycle tracking

4. USER ACCESS & AUTHENTICATION

4.1 Authentication Mechanism

The system uses secure authentication mechanisms to protect device access and user data. **Security Features:**

- JWT-based session authentication
- Bcrypt password hashing
- Token validation middleware
- Device-level access filtering • Secure FCM token mapping

4.2 User Types

The system operates with two primary user types:

Admin

Admin has complete system access.

Admin can:

- Register and manage devices
- Create and manage user accounts
- Monitor all borewell devices
- View system-wide telemetry and history
- Manage shop products and pricing
- Monitor and manage customer orders
- Activate or deactivate devices

User

User is the borewell motor owner or operator.

User can:

- Start / Stop motor remotely
- Monitor live voltage, current, and power
- Receive alerts (Dry run, overload, etc.)
- View motor session history
- Track daily energy consumption
- Purchase products
- View order status
- Update profile information

5. COMMUNICATION PROTOCOL

5.1 Protocol Details

Parameter	Value
Protocol	MQTT v3.1.1
QoS	1
Data Format	JSON
Transport	TCP over GSM/4G
Keep Alive	60 seconds

5.2 Topic Structure

All MQTT communication follows the topic pattern:

`agri/{serial_number}/{message_type}`

Where:

- {serial_number} = Unique device identifier
- {message_type} = boot / heartbeat / telemetry / status / alert / cmd

5.3 Boot Message (Device to Cloud)

Topic to Publish: agri/{serial_number}/boot

Description:

Sent once when the device power on or reconnects to the network.

Payload Format:

```
{
  "MESSAGE_TYPE": "BOOTNOTIFICATION",
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "FIRMWARE_VERSION": "1.0.3",
  "HARDWARE_VERSION": "HW1.0",
  "TIMESTAMP": "ISO-8601-STRING",
  "POWER_STATUS": "ON",
  "NETWORK_STATUS": "4G",
}
```

5.4 Heartbeat Message (Device to Cloud)

Topic to Publish: agri/{serial_number}/heartbeat

Description:

Sent periodically (every 15 seconds) to indicate the device is online. If device is idle.

Payload Format:

```
{
  "MESSAGE_TYPE": "HEARTBEAT",
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "SIGNAL_STRENGTH": 80,
  "TIMESTAMP": "ISO-8601-STRING",
  "MOTOR_RUNNING": true
}
```

Purpose:

- Monitor device health
- Detect online/offline state
- Track signal quality

5.5 Telemetry Message (Device to Cloud)

Topic to Publish: `agri/{serial_number}/telemetry`

Description:

Primary real-time monitoring message sent while motor is running.

Sent this telemetry data to cloud every 1 Minutes.

Payload Format:

```
{
  "TYPE": "TELEMETRY",
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "TIMESTAMP": "ISO-8601-STRING",
  "VOLTAGE_RMS": 230.5,
  "CURRENT_RMS": 4.5,
  "MOTOR_RPM": 2100,
  "POWER_KW": 1.1,
  "ENERGY_KWH": 0.05,
  "DEVICE_TEMP_C": 38.5,
  "FLOW_LPM": 250,
  "FREQUENCY_HZ": 50,
  "POWER_FACTOR": 0.92,
  "SIGNAL_STRENGTH": 75,
  "FAULT_CODE": 0
}
```

Processing Steps:

- Store telemetry data in database
- Update device state
- Calculate energy consumption
- Detect fault conditions
- Emit LIVE_TELEMETRY event via Socket.io
- Trigger alert notification if "FAULT_CODE": 1 -> If any fault occur at the time send this fault code "1" at the same time device need to send an alert frame to cloud.

5.6 Status Message (Device to Cloud)

Topic to Publish: agri/{serial_number}/phase

Description:

Sent after executing Start/Stop command.

Payload Format:

```
{  
  "MESSAGE_TYPE": "STATUS",  
  "SERIAL_NUMBER": "SN12345",  
  "IMEI_NUMBER": "8645...",  
  ""MOTOR_RUNNING": true  
  "MOTOR_POWER": "Single Phase",  
  "TIMESTAMP": "ISO-8601-STRING"  
}
```

MOTOR_POWER: Single Phase /Two Phase/ Three Phase

MOTOR_STATUS: RUNNING / STOP

Every one minute need to send to cloud.

5.7 Alert Message (Device to Cloud)

Topic to Publish: agri/{serial_number}/alert

Description:

Sent immediately when a fault or abnormal condition is detected.

Payload Format:

```
{  
  "MESSAGE_TYPE": "ALERT",  
  "ALERT_TYPE": "DRY_RUN",  
  "SERIAL_NUMBER": "SN12345",  
  "IMEI_NUMBER": "8645...",  
  "DEVICE_STATUS": "CRITICAL",  
  "FAULT_CODE": 101,  
  "TIMESTAMP": "ISO-8601-STRING"  
}
```

Types Of Alerts: DRY_RUN - 101, OVER_VGE - 102/OVER_I- 103, OVER_LOAD - 104,
BCKWRD_RUN -105

Purpose:

- Log fault in alerts collection
- Send instant push notification
- Update device status to warning/critical

5.8 Command Message (Cloud → Device)

Topic to Subscribe: `agri/{serial_number}/cmd`

Description:

Used by backend to send motor control commands.

Payload Format:

```
{
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "COMMAND": "START_MOTOR",
  "USER_ID": 1,
  "TIMESTAMP": "ISO-8601-STRING"
}
```

Possible COMMAND values:

- START_MOTOR
- STOP_MOTOR

Processing Flow:

1. Backend validates user permission
2. Publish MQTT command
3. Device executes command
4. Device sends STATUS acknowledgment

5.9 Command Message (Cloud → Device)

Topic to Subscribe: `agri/{serial_number}/cmd`

Description:

Used by backend to send motor control commands (Read Configurations)

Payload Format:

```
{
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "COMMAND": "READ_CONFIG",
  "USER_ID": 1,
  "TIMESTAMP": "ISO-8601-STRING"
}
```

Possible COMMAND values:

- READ_CONFIG

5.10 Command Message (Device → Cloud)

Topic to Publish: `agri/{serial_number}/resp`

Description:

Used by device to send current configuration details to backend. (Read Configurations)

```
{
  "SERIAL_NUMBER": "SN12345",
  "COMMAND": "READ_CONFIG",
  "STATUS": "SUCCESS",
  "CONFIG": {
    "MOTOR_DEVICE_ID": "MD-001",
    "FIRMWARE_VERSION": "1.2.0",
    "MAX_VOLTAGE": 260,
    "MIN_VOLTAGE": 180,
    "MAX_CURRENT": 12,
    "MIN_CURRENT": 2,
    "POWER_RATING": "5HP"
  },
  "TIMESTAMP": "ISO-8601-STRING"
}
```

Possible STATUS values:

- SUCCESS
- FAILED

5.11 Command Message (Cloud → Device)

Topic to Subscribe: `agri/{serial_number}/cmd`

Description:

Used by device to send current configuration details to backend. (Set Configurations)

```
{
  "SERIAL_NUMBER": "SN12345",
  "COMMAND": "SET_CONFIG",
  "STATUS": "SUCCESS",
  "CONFIG": {
    "MOTOR_DEVICE_ID": "MD-001",
    "FIRMWARE_VERSION": "1.2.0",
    "MAX_VOLTAGE": 260,
    "MIN_VOLTAGE": 180,
    "MAX_CURRENT": 12,
    "MIN_CURRENT": 2,

```

```
    "POWER_RATING": "5HP"
  },
  "TIMESTAMP": "ISO-8601-STRING"
}
```

Possible COMMAND values:

- SET_CONFIG

5.12 Command Message (Device → Cloud)

Topic to Publish: agri/{serial_number}/resp

Description:

Used by device to send current configuration details to backend. (Read Configurations)

```
{
  "SERIAL_NUMBER": "SN12345",
  "COMMAND": "SET_CONFIG",
  "STATUS": "SUCCESS",
  "TIMESTAMP": "ISO-8601-STRING"
}
```

Possible STATUS values:

- SUCCESS
- FAILED

5.13 Command Message (Cloud → Device)

Topic to Subscribe: agri/{serial_number}/cmd

Description:

Used by backend to initiate firmware Over-The-Air (OTA) update on the device.

```
{
  "SERIAL_NUMBER": "SN12345",
  "IMEI_NUMBER": "8645...",
  "COMMAND": "OTA_UPDATE",
  "FW_VERSION": "1.3.0",
  "FW_URL": "https://server.com/firmware.bin",
}
```

5.14 Command Message (Cloud → Device)

Topic to Publish: `agri/{serial_number}/resp`

Description:

Used by device to report OTA progress and final status. Before starting the OTA, send the Current status to Cloud like in-Progress, Rejected. After completing the OTA send Success message.

```
{  
  "SERIAL_NUMBER": "SN12345",  
  "COMMAND": "OTA_UPDATE",  
  "STATUS": "IN_PROGRESS",  
  "TIMESTAMP": "ISO-8601-STRING"  
}
```

Possible STATUS values:

- IN_PROGRESS
- SUCCESS
- FAILED
- REJECTED

6. CONCLUSION

AgriPlus is a scalable and secure IoT-based Borewell Motor Automation platform designed for modern agriculture.

The system provides:

- Real-time motor monitoring
- Secure remote control
- Intelligent fault detection
- Energy tracking
- Integrated e-commerce

With strong MQTT-based telemetry processing and secure role-based access control, AgriPlus is designed for reliable large-scale deployment.

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